

1 Number

What students need to learn:

Ref	Content descriptor	Concepts and skills
N a	Add, subtract, multiply and divide any number	• Add, subtract, multiply and divide whole numbers, integers, negative numbers, decimals, fractions and numbers in index form • Recall all multiplication facts to 10×10, and use them to derive quickly the corresponding division facts • Multiply or divide any number by powers of 10 • Multiply or divide by any number between 0 and 1 • Solve a problem involving division by a decimal (up to two decimal places) • Write numbers in words • Write numbers from words • Recall the fraction-to-decimal conversion of familiar fractions
N b	Order rational numbers	• Order integers, decimals and fractions • Understand and use positive numbers and negative integers, both as positions and translations on a number line
N c	Use the concepts and vocabulary of factor (divisor), multiple, common factor, Highest Common Factor (HCF), Least Common Multiple (LCM), prime number and prime factor decomposition	• Recognise even and odd numbers • Identify factors, multiples and prime numbers • Find the prime factor decomposition of positive integers • Find the common factors and common multiples of two numbers • Find the Lowest common multiple (LCM) and Highest common factor (HCF) of two numbers

Ref	Content descriptor	Concepts and skills
N d	Use the terms square, positive and negative square root, cube and cube root	- Recall integer squares up to 15×15 and the corresponding square roots - Recall the cubes of 2, 3, 4, 5 and 10 - Find squares and cubes - Find square roots and cube roots
N e	Use index notation for squares, cubes and powers of 10	- Use index notation for squares and cubes - Use index notation for powers of 10 - Find the value of calculations using indices
N f	Use index laws for multiplication and division of integer powers	Use index laws to simplify and calculate the value of numerical expressions involving multiplication and division of integer powers, and of powers of a power (NB: Fractional, zero and negative powers are only assessed on Higher Tier)
N h	Understand equivalent fractions, simplifying a fraction by cancelling all common factors	- Find equivalent fractions - Write a fraction in its simplest form - Convert between mixed numbers and improper fractions - Compare fractions
N i	Add and subtract fractions	Add and subtract fractions
N j	Use decimal notation and recognise that each terminating decimal is a fraction	- Understand place value - Identify the value of digits in a decimal or whole number - Write terminating decimals as fractions - Recall the fraction-to-decimal conversion of familiar simple fractions - Convert between fractions and decimals
N k	Recognise that recurring decimals are exact fractions, and that some exact fractions are recurring decimals	- Recognise that recurring decimals are exact fractions, and that some exact fractions are recurring decimals - Convert between recurring fractions and decimals

Foundation

Ref	Content descriptor	Concepts and skills
N i	Understand that 'percentage' means 'number of parts per 100' and use this to compare proportions	• Order fractions, decimals and percentages • Convert between fractions, decimals and percentages
N m	Use percentage	• Use percentages to solve problems • Find a percentage of a quantity in order to increase or decrease • Use percentages in real-life situations – VAT – Value of profit or loss – Simple Interest – Income tax calculations (NB: Repeated proportional change is only assessed at Higher Tier)
N o	Interpret fractions, decimals and percentages as operators	• Find a fraction of a quantity • Express a given number as a fraction of another • Find a percentage of a quantity • Express a given number as a percentage of another number • Use decimals to find quantities • Understand the multiplicative nature of percentages as operators • Use a multiplier to increase or decrease by a percentage in any scenario where percentages are used
N p	Use ratio notation, including reduction to its simplest form and its various links to fraction notation	• Use ratios • Write ratios in their simplest form

Ref	Content descriptor	Concepts and skills
N q	Understand and use number operations and the relationships between them, including inverse operations and hierarchy of operations	• Multiply and divide numbers using the commutative, associative, and distributive laws and factorisation where possible, or place value adjustments • Use inverse operations • Use brackets and the hierarchy of operations • Use one calculation to find the answer to another • Understand 'reciprocal' as multiplicative inverse, knowing that any non-zero number multiplied by its reciprocal is 1 (and that zero has no reciprocal because division by zero is not defined) • Find reciprocals • Understand and use unit fractions as multiplicative inverses • Solve word problems
N t	Divide a quantity in a given ratio	• Divide a quantity in a given ratio • Solve a ratio problem in context
N u	Approximate to specified or appropriate degrees of accuracy including a given power of ten, number of decimal places and significant figures	• Round numbers to a given power of 10 • Round to the nearest integer and to any given number of significant figures • Round to a given number of decimal places • Estimate answers to calculations, including use of rounding
N v	Use calculators effectively and efficiently, including statistical functions	• Know how to enter complex calculations • Enter a range of calculations including those involving time and money • Understand and interpret the calculator display, knowing when the display has been rounded by the calculator, and know not to round during the intermediate steps of a calculation • Use a range of calculator functions including $+$, $-$, $\times$, $\div$, x^2, $\sqrt{x}$, memory, x^y, $x^{1/y}$ and brackets